2 **(a)** $F = M_x D_x \omega_x^2 = M_y D_y \omega_y^2$

$(M)(2D)\ \omega_x^2 = (2M)(D)\ \omega_y^2$ [1]

$\omega_x = \omega_y$

(b) Gravitation force provides centripetal force

$GM(2M)/(3D)^2 = (2M)D\omega^2$

$\omega^2 = GM/(9D^3)$

$KE_x = ½\ M(2D\ \omega)^2 = 2GM^2/(9D)$ [1]

$KE_y = ½\ (2M)(D\ \omega)^2 = GM^2/(9D)$ [1]

$GPE = -\ GM(2M)/(3D)$ [1]

$E = 2GM^2/(9D) + GM^2/(9D) - GM(2M)/(3D)$

$E = -\ GM^2/(3D)$ [1]

(c) A negative total energy signifies that the system is gravitationally bound [1]

Or

The stars lack sufficient energy to overcome their mutual attraction and escape to an infinite separation.

(d) Not possible since force between identical charges is repulsive and will not point towards the centre of motion. [1]